

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	1	of	27

NSP/004/104 (OHI 4) Guidance on the Types and Installation Requirements for stays

1. Purpose

The purpose of this document is to provide guidance on the types and installation requirements for stays on all wood pole constructions used on the CE Electric UK distribution system.

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/104	2.0	Guidance on the types and installation requirements for stays

2. Scope

This document provides guidance on the types, rating and installation requirements for stay assemblies and their individual components. This guidance relates to all voltages up to and including 132kV.

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	2	of	27

2.1 Contents

1.	Purpose	1
2.	Scope.....	1
2.1	Contents.....	2
3.	Technical Specification	3
3.1	Background.....	3
3.2	Stay arrangements	3
3.2.1	Strut Poles.....	3
3.3	Stay Wire.....	3
3.4	Stay Blocks & Rods.....	4
3.5	Stay Rods Terminations	4
3.6	Pole Top Termination's.....	4
3.6.1	Pole top stay dead end.....	4
3.6.2	Pole top stay make off.....	5
3.6.3	Stay Wire Splice	5
3.6.4	Stay Plates	5
3.7	Stay insulator terminations	6
3.8	Stay Insulators	6
3.8	Stay Fencing.....	6
3.9	Stay Wire Guards	7
3.10	Stay setting out and stay positions.....	7
3.11	Temporary Stays.....	7
3.12	Alternative stay anchoring systems.....	7
3.12.1	Duckbill (Platipus) Stay Anchors.....	7
3.12.2	Augured Stay Anchors	8
3.13	Stay Charts	8
3.14	Clearances.....	8
4.	References.....	10
4.1	External Documentation	10
4.2	Internal documentation	10
4.3	Amendments from Previous Version	10
5.	Definitions	10
6.	Authority for issue	11
6.1	Author.....	11
6.2	Technical Assurance.....	11
6.3	DBD Assurance	11
6.4	Approval.....	11
6.5	Authorisation.....	11
	Appendix 1 – LV Stay Arrangement.....	12
	Appendix 2 - 11kV to 33kV Typical Stay Arrangement.....	14
	Appendix 3 – 11kV to 33kV Alternative Stay Arrangement.....	16
	Appendix 4 - Typical CE/OM/DC1 Stay Arrangement	17
	Appendix 5 – Typical CE/C/37 Stay Arrangement.....	18
	Appendix 5 – Typical CE/C/37 Stay Arrangement.....	19
	Appendix 6 – 66kV/132kV Typical Arrangement	21
	Appendix 7 – Stay Make offs	23
	Appendix 8 - Platipus Utility System	24
	Appendix 9 – Installation Instructions for Augured Ground Anchor Systems.....	26

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	3	of	27

3. Technical Specification

3.1 Background

Stays are fitted to poles to counteract conductor tension forces and restrain deflection of the pole top into the conductor span, thereby maintaining poles PLUMB, at angle and terminal positions when conductors are erected. To allow for this the pole should be erected leaning slightly towards the stays when initially made off with the threaded shank of the stay rod inserted not less than 40 mm into the stay barrel.

The optimum angle between the pole and the stay should be 45°, but in practice this is not always possible. Design specifications must be reviewed to confirm that stays are allowed at lesser angles. In all cases the minimum stay angle shall be 30° between stay and pole. A range of stay charts have been produced for different line specifications detailing permissible stay angles below 45°. See clause 3.13 for further details.

3.2 Stay arrangements

The general arrangement of stays for the different types of construction shall be as follows:

Construction	Drawing Arrangement
LV Construction (NSP/004/041)	1091193326
HV Construction (NSP/004/042)	1000439108
33/66kV Construction CE/C/37 & CE/OM/DC1 (DSS/004/040)	Y204L0501 or Y205L0501
66/132kV Construction OHL 4 / OHL 9 (Arrangement drawing 1091231192 requires the use of 439113 stay straps)	1091231166

3.2.1 Strut Poles

Under exceptional circumstances where no other traditional stayed support can be utilised due to wayleave or access problems, a strut pole arrangement as shown on the following drawings may be utilised. The use of this arrangement is limited to a maximum conductor CSA of 50mm AAAC on HV Lines.

Voltage	Conductor CSA	Max Angle of Deviation	Drawing Arrangement
11 - 20kV	50mm AAAC	30°	1000434001 sht 2
LV	All conductor Sizes	30°	1091193331

3.3 Stay Wire

- Stay wire is manufactured from 7 strands of 4.00mm diameter grade 1150 grade galvanised wire (7/4.00), made up with a right hand lay. The stay wire shall have a minimum failing load of 101 kN.
- This grade matches the grade used in the manufacture of helical stay fittings.
- Steel of this quality is very hard, making it very difficult to cut and impossible to hand splice. For this reason all stays are made off using helical stay fittings which are suitable for wrapping onto right hand lay stay wire.

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	4	of	27

3.4 Stay Blocks & Rods

All stay shall be installed utilising 439103 types 2 stay blocks installed to a depth to the bottom of the block, as per the table below.

Voltage	Stay at 45° to pole	Stay at 30° to pole
LV	1.6m	1.8m
HV & EHV	1.8m	2.0m

In all cases the block must be installed into undisturbed ground to provide maximum resistance to uplift. A slot must be cut into the side of the stay hole so that the stay rod is in line with the stay wire. See NSP/004/101 for further details.

A common design of adjustable stay rod rated at 110kN shall be used for all voltages See drawing number 1000439101 sheet 2 for details.

In ground with a high acid content, such as peat or refuse tip sites, it is necessary to wrap the rod with two layers of grease impregnated tape (Denso), applied at 50% overlap, in order to protect the stay rod against corrosion.

3.5 Stay Rods Terminations

Stays shall be terminated onto the stay rods using helical dead end stay grips or hand wrapped GPO make offs. Detailed installation instructions for this termination can be found in WI 74/10.

Where a preformed make off is used, the stay wire shall be cut so that there is no protrusion of the wire through the stay perform loop that could cause injury to animals rubbing against the stay.

3.6 Pole Top Termination's

The following methods of pole top stay termination may be used:

- Pole top dead end to drawing no 1000439108 sheet 6
- Pole top stay make off
- Stay Plates
- Stay Straps (OHL 4 Construction) or Stay Bracket (CE/C/37)
- Hand wrapped GPO to drawing no 1091010181

3.6.1 Pole top stay dead end

- A pole top stay dead is a galvanised steel helical fitting used to terminate a length of 7/4.00mm stay strand. The other end of the stay strand shall then be terminated directly into a guy grip dead end containing the required insulator type. (See notes on positioning of stay insulators).
- Where the stay is required to be bonded to the pole top crossarm this shall be achieved by cutting the outer strands of the offered stay strand in such a way that it leaves sufficient length of "kingwire" to reach onto the structure earthing point.
- Application instructions for pole top dead ends can be found in work spec 04/020.

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	5	of	27

3.6.2 Pole top stay make off

- A pole top stay make off is a galvanised steel helical fitting that when applied at the pole top, the completed tail ends form into a 1.8m length of 7/4.00mm stay strand.
- The newly formed stay strand tail shall be terminated directly into a guy grip dead end containing the required insulator type. (see notes on positioning of stay insulators)
- Where it is necessary to locate a stay insulator at a height lower than 1.8m from the pole top, this shall be achieved by extending the tail from the pole top stay make off with a “stay wire splice” and a short length of 7/4.00mm stay strand. The extended length stay may then be terminated onto a stay insulator in a similar way to the pole top dead end described above.
- A medium tensile quality single bond wire is incorporated into this fitting for bonding purposes.
- See WI 74-13 for application guidance.

3.6.3 Stay Wire Splice

- A stay wire splice is a 1.4m long galvanised steel helical fitting that can be utilised to joint/extend 7/4.00mm stay strand.
- See WI 74/12 for application guidance

3.6.4 Stay Plates

Drawing for Stay Plate/Straps	Cat Number	Stay Wire Suitability
1000439110 (70kN MFL) - Single	237134	Single 4/7.00mm grade 700 stay strand (Only suitable for LV stays or historical HV stays)
1091010207 (125kN MFL) - Double	242005	Single 4/7.00mm grade 1150 stay strand or Twin 4/7.00mm grade 700 stay strand
Stay Bracket – Y204L0503 (CE/C/37 Lines)	220343	Single 4/7.00mm grade 1150 stay strand
1000439113 (190kN MFL) – OHL 4 Lines	252647	Single 4/7.00mm grade 1150 stay strand or Twin 4/7.00mm grade 1150 stay strand

- Stay plates to 1000439110 are only suitable for use on LV lines.
- Where an angle pole requires a single stay (grade 1150) the stay plate shall be fitted as shown on drawing no 1000439108 sheet 4 fig1.
- Where two grade 1150 stays are required at an angle pole the stay plates shall be fitted as shown on drawing no 1000439108 sheet 4 fig 2.
- Stay plates may be used to terminate single grade 1150 stays per structure leg. See drawing 1000439108 sheet 5 fig1, for details. Where two stays per leg are required, that shall be terminated in accordance with clauses 3.6.1 or 3.6.2 of this document.

Note: In all cases where a stay plate is used, the M20 bolt should be fitted such that the bolt head is retaining the stay plate with the square curved washer and full nut on the opposite side of the pole.

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	6	of	27

3.7 Stay insulator terminations

Single insulators in a stay shall be terminated using helical guy grip dead ends as shown on drawing 1000439107 sheet 2. (See WI 74/11 for application guidance).

Twin insulators in a stay shall be terminated using helical stay link assemblies as shown on drawing 1000439107 sheet 3. (See WI 74/14 for application guidance).

3.8 Stay Insulators

Every stay shall be:-

- Fitted with an approved type of insulator(s) positioned as specified on the general arrangement drawing.
- Bonded at the top, to the line crossarm.

Note – Stay Insulator Positioning

In situations where a broken jumper or a broken stay may result in a stay becoming 'live' the insulator shall be placed below any likely point of contact with live metal, but NOT below a position which would maintain a minimum of 3.0m above the ground when the stay swings vertical if broken. The most common circumstances for this situation is on poles carrying auxiliary apparatus and the stay insulator(s) should be repositioned from the standard 1.8m or 3m from the pole top to ensure that this hazard will not occur.

Approved insulators are shown below

Voltage	No. Req.	Insulator Type or Drawing No.
LV poles	1	1000439107 Type 1
All earthed poles	1	1000439107 Type 1
11kV unearthed poles	1	1000439107 Type 1 * see below
20kV unearthed poles	2	1000439107 Type 1
33kV unearthed poles	2	1000439107 Type 2
66/132kV unearthed poles	1	1091010372 Sheet 2

* Except in the old NEDL North Yorkshire District where 2 type 1 stay insulators shall be used with the link assembly. This policy may also be applied in other areas where evidence is found of lightning flashover on the stay insulators (System should be designed to encourage flashover at triggered spark gaps).

See WI 74/14 for application guidance on the insulator link assembly

Stay insulators of the glass fibre rod type manufactured previously by CCL should be replaced during dead line maintenance. Care should be taken when installing insulators in previously un-insulated stays as they were prone to corrosion of the stay rod due to circulating leakage currents. The stay rod should be inspected before stay wire replacement or insulator insertion.

3.8 Stay Fencing

Stay fencing shall be fitted as shown on drawing no 1091010099 as determined by the stay location, use of the land and the direction of the land owner/wayleave officer.

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	7	of	27

3.9 Stay Wire Guards

Stays at the edge of footpaths or in hedges cut by hedge cutting machines shall have yellow PVC stay guards fitted (cat 261970) to give visual indication as to the presence of the stay wire.

3.10 Stay setting out and stay positions

- Stay positions at angle poles shall be set out on site in accordance with NSP/004/101.
- On lay terminal tee-off arrangements with J&P fuse isolators on the main line pole, the stay positions should be set out to achieve the maximum stay angle whilst still ensuring that the stay to jumper clearances specified in NSP/004/107 are not infringed.

3.11 Temporary Stays

To facilitate conductor running and pre-stressing of conductors over a number of line sections it is necessary to install temporary stays at section positions.

3.12 Alternative stay anchoring systems

3.12.1 Duckbill (Platipus) Stay Anchors

The duckbill stay anchor system provides an alternative ground anchor to the traditional wood block and stay rod. The system utilises portable compressors to drive specially shaped anchors into the ground. Once installed the attached stay tendon is pulled back out of the ground using the accompanying load lock jack. This process turns the anchor head around through 90° resulting in the anchor becoming fixed in the ground. The usual above ground stay components can then be attached to the fully proof tested ground anchor.

An excavator still remains the most economical means of installing a stay rod and block. However the duckbill system only comes into its own where an excavator is not available on site or where stays are required in sensitive hedge lines where excavations may damage the existing fauna.

Each installed anchor must be proof tested with the load lock jack to the criteria below and the stay installed in accordance with the minimum requirements stay chart requirements. The minimum separation distance between any two duckbill stay anchors will be one metre between tendons.

Drawing no 1091010024 sheet 1 and 2 provides details on the installation equipment.

Full installation instruction can be found in Appendix 1

The effectiveness of this ground anchor system is dependant upon the ground type where it is installed. If the required proof load cannot be achieved, the installation of the anchor system shall be replaced with the traditional rod and block system.

Situation	Min Proof Load Requirement
All LV overhead lines with a minimum proof load	25kN
Three phase & single phase HV overhead lines with conductor CSA up to 100mm diameter.	30kN
Three phase & single phase HV overhead lines with conductor	

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	8	of	27

CSA up to 175mm diameter.	35kN
For use on Out of balance Stays – any conductor size	30kN

3.12.2 Augured Stay Anchors

The AB Chance screw anchor system provides an alternative ground anchor to the traditional wood block and stay rod system. This system utilises a JCB or Unimog to mount a hydraulically driven Auger motor. The Auger motor provides a rotational drive system to power the helix screw anchors into the ground.

A documented relationship exists between the torque required to drive the anchor into the ground and its resultant ground holding capability.

The installed anchor can then be terminated to the stay wire in the usual way using helical preformed fittings. Detailed instructions are included in Appendix 9 of this document.

3.13 Stay Charts

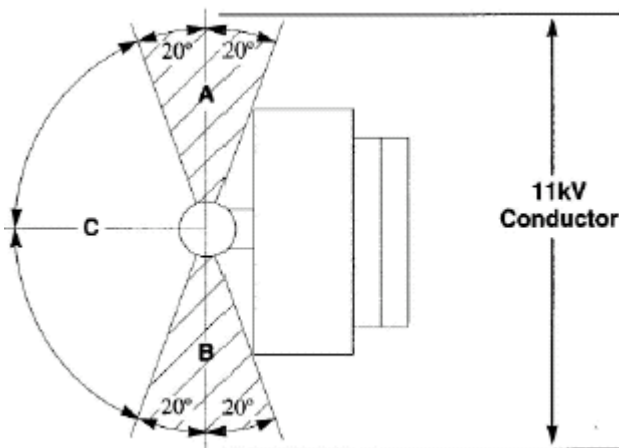
The following documents provide sag charts detailing the minimum allowed stay angles or stay spreads for different design specifications

Specification	DBD doc ref	Appendix Number
Specification for LV ABC Lines	NSP/004/041	Appendix 9 & 10
Renovation of existing LV overhead lines	NSP/004/041/001	Appendix 10
Specification for HV lines up to 33kV	NSP/004/042	Appendix 2 – 11 (or 25 – 43 for renovation work)
Specification for covered conductor HV lines up to 33kV	NSP/004/044	Appendix 2 – 6

3.14 Clearances

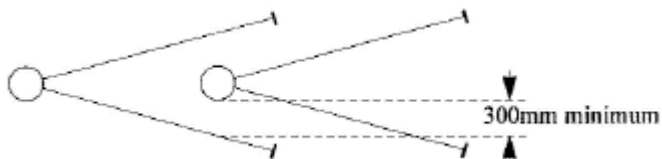
Where a low voltage line is attached to a pole at the same level as or above a transformer on the pole:

- Conductors in tension and backstays are only permitted in the shaded area (A & B), shown on the diagram below.
- Low voltage slack spans may be located anywhere in the shaded or unshaded sectors (A & B or C), provided that a minimum clearance of 300mm is maintained from any staywire.



Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	9	of	27

- When a low voltage overhead line is attached to a pole below the level of the transformer on that pole, no restriction to the direction of slack spans, tension spans or stays applies.
- Where 'H' supports are used at angle positions it is necessary to make sure that 300mm clearance exists between stay and pole surface, at the point where the stay wire attached to the inner limb passes the outer limb, as shown in the diagram below.



Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	10	of	27

4. References

4.1 External Documentation

Reference	Title
-----------	-------

4.2 Internal documentation

Reference	Title
WI 74-10	Work Instruction for Application of a Helical Dead End Stay Grip to a Stay Rod
WI 74-11	Work Instruction for Application of a Helical Dead End Stay Grip to a Stay Insulator
WI 74-12	Work Instruction for Application of a Helical Stay Splice
WI 74-13	Work Instruction for Application of a Helical Pole Top Wrap (Pole Top Make Off)
WI 74-14	Work Instruction for Application of a Helical Insulator Link Assembly
04/020	NEDL Work Instruction for the application of pole top dead ends

4.3 Amendments from Previous Version

Clause	Subject	Amendments
2.1	Contents	Added to document
4.3	Amendments from Previous Version	Added to document
Appendix 9	Installation Instructions for Augured Ground Anchor Systems	Added to document

5. Definitions

Term	Definition
------	------------

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	11	of	27

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager	G Hammel	28/05/2010

6.2 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
S Salkeld	Standards Engineer	S Salkeld	28/05/2010

6.3 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	17/06/2010

6.4 Approval

Approval is given for the content of this document

		Sign	Date
A Bilclough	Overhead Program Manager	A Bilclough	14/06/2010

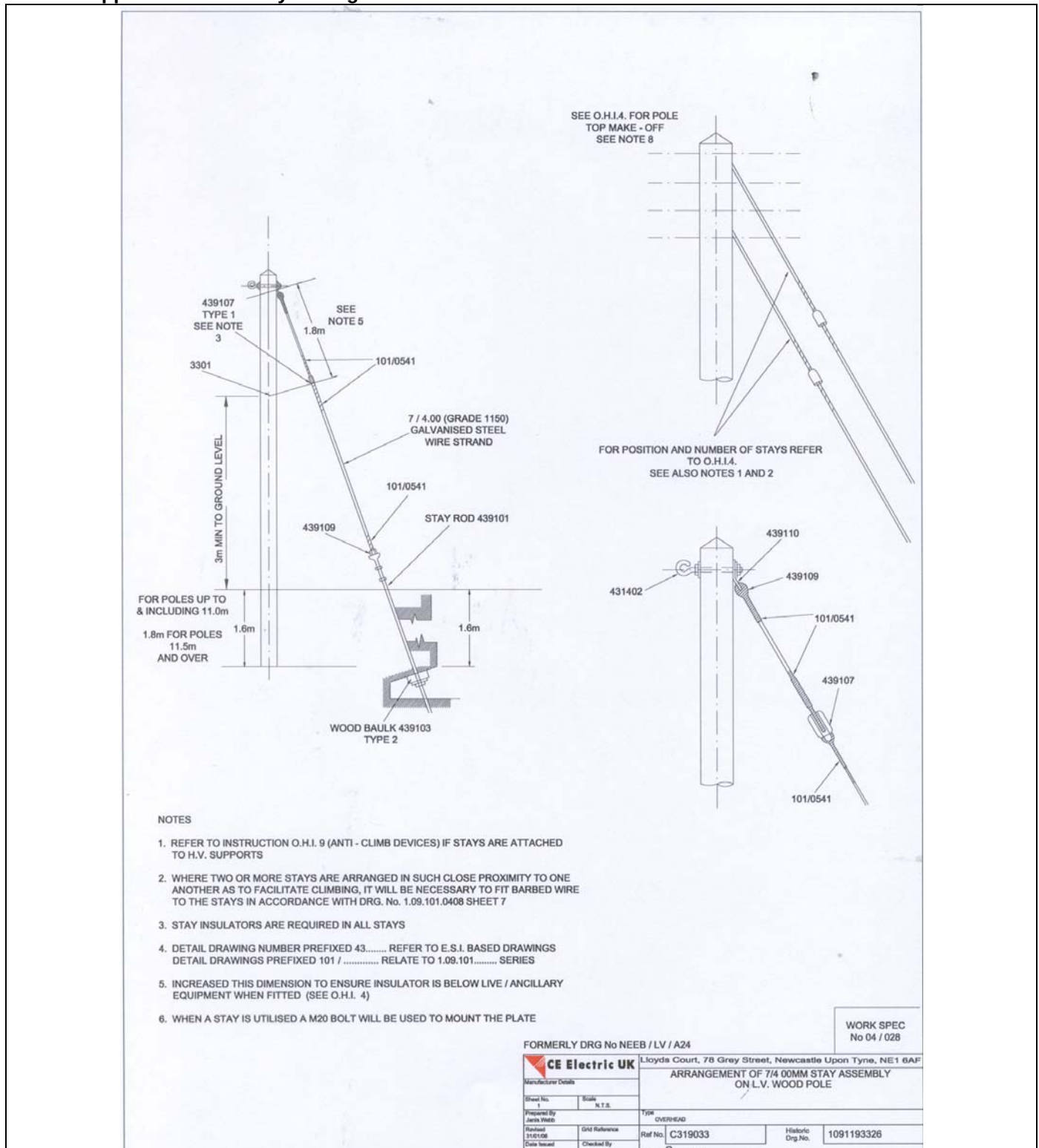
6.5 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
S Caskey	Head of Program Delivery	S Caskey	08/06/2010

Document reference		NSP/004/104						
Version:-	3.0	Date of Issue:-	June 2010	Page	12	of	27	

Appendix 1 – LV Stay Arrangement



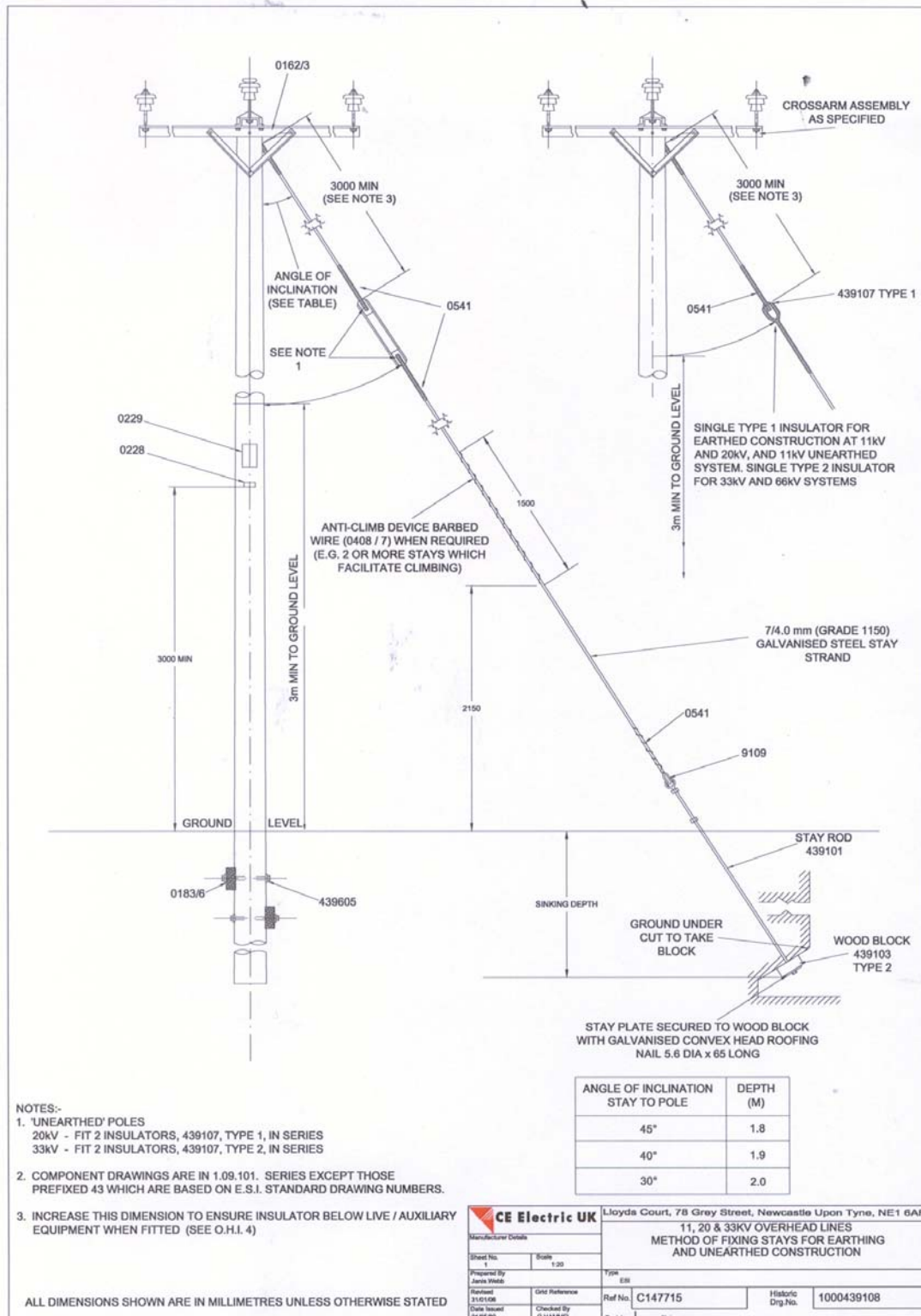
Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	13	of	27

Materials for 1000439108 sht1

Item	No	Cat	Description	Drawing
16	1	235124	Kicking /Stay Block (Type2)	1000439103
100	1	231477	Stay Rod (Type 2)	1000439101
101		229583	Stay Wire (7/4.00mm) Grade 1150	
63	1	248232	Stay Insulator (Type 1)	1000439107
-	1	237134	Stay Plate	1000439110
102	4	255217	Stay Grip Dead End	1091010541

Document reference	NSP/004/104						
Version:-	3.0	Date of Issue:-	June 2010	Page	14	of	27

Appendix 2 - 11kV to 33kV Typical Stay Arrangement



Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	15	of	27

Materials for 1000439108 sht1

Item	No	Cat	Description	Drawing
16	1	235124	Kicking /Stay Block (Type2)	1000439103
100	1	231477	Stay Rod (Type 2)	1000439101
101		229583	Stay Wire (7/4.00mm) Grade 1150	
102	3	255217	Stay Grip	1091010541
-	3	244922	Stay Thimble	1000439109
Pole Top Terminations				
105	1	225342	Pole top make off	
104	1	254089	Stay Wire Splice	
101		229583	Stay Wire (7/4.00mm) Grade 1150	
Or				
-	1	250209	Pole top dead end	1000439108 sht6
101		229583	Stay Wire (7/4.00mm) Grade 1150	
Or				
102	3	255217	Stay Grip	1091010541
-	1	237134	Stay Plate	1000439110

Insulator requirements for:-

11, 20, 33, 66 or 132kV unearthed arrangements

63	1	248232	Stay Insulator (Type 1)	1000439107
----	---	--------	-------------------------	------------

20kV Unearthed Arrangement

63	2	248232	Stay Insulator (Type 1)	1000439107
103	1	260183	Stay Link Assembly	1000439107 sht3

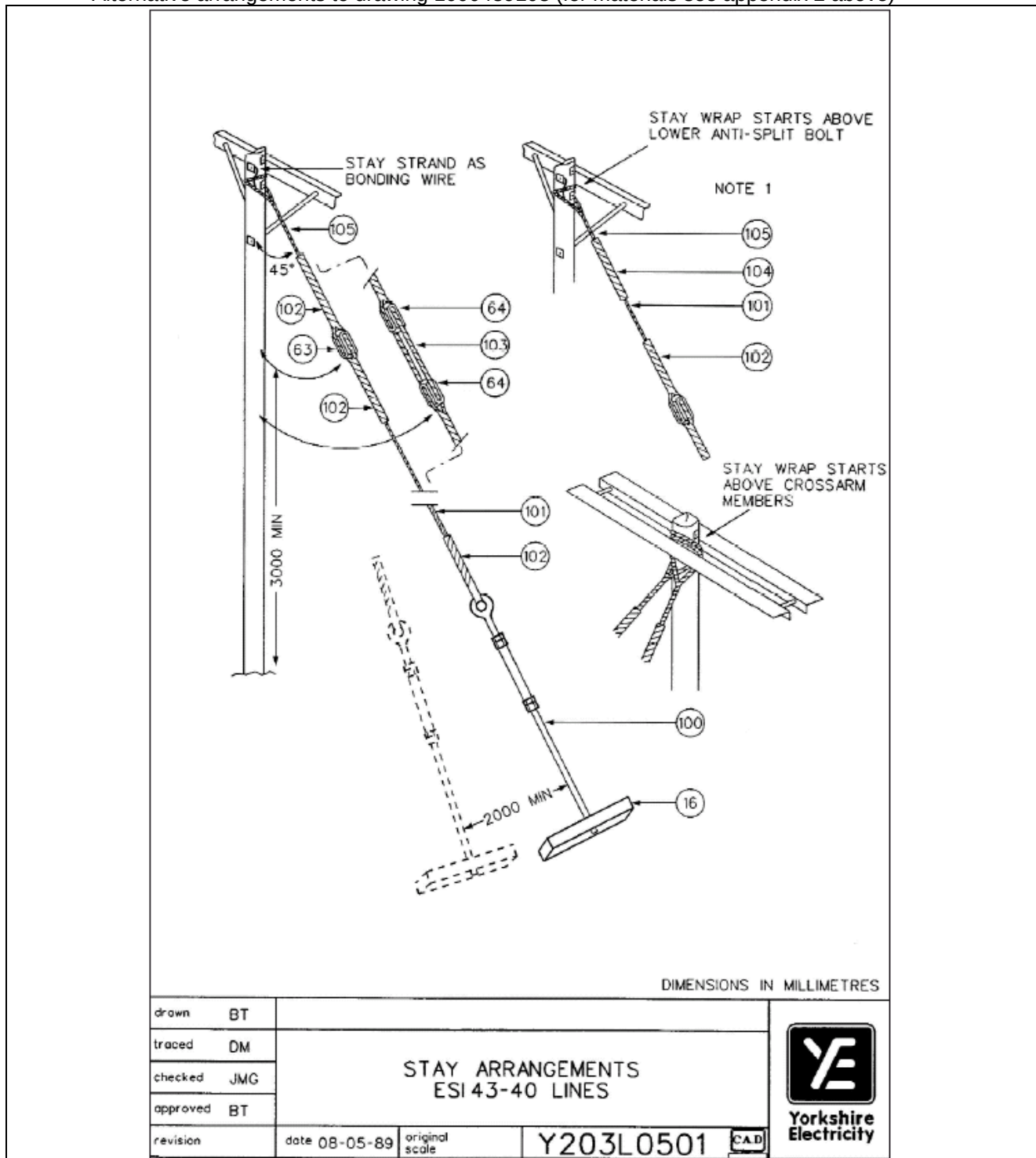
33kV Unearthed Arrangement

63	2	253743	Stay Insulator (Type 2)	1000439107
103	1	260183	Stay Link Assembly	1000439107 sht3

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	16	of	27

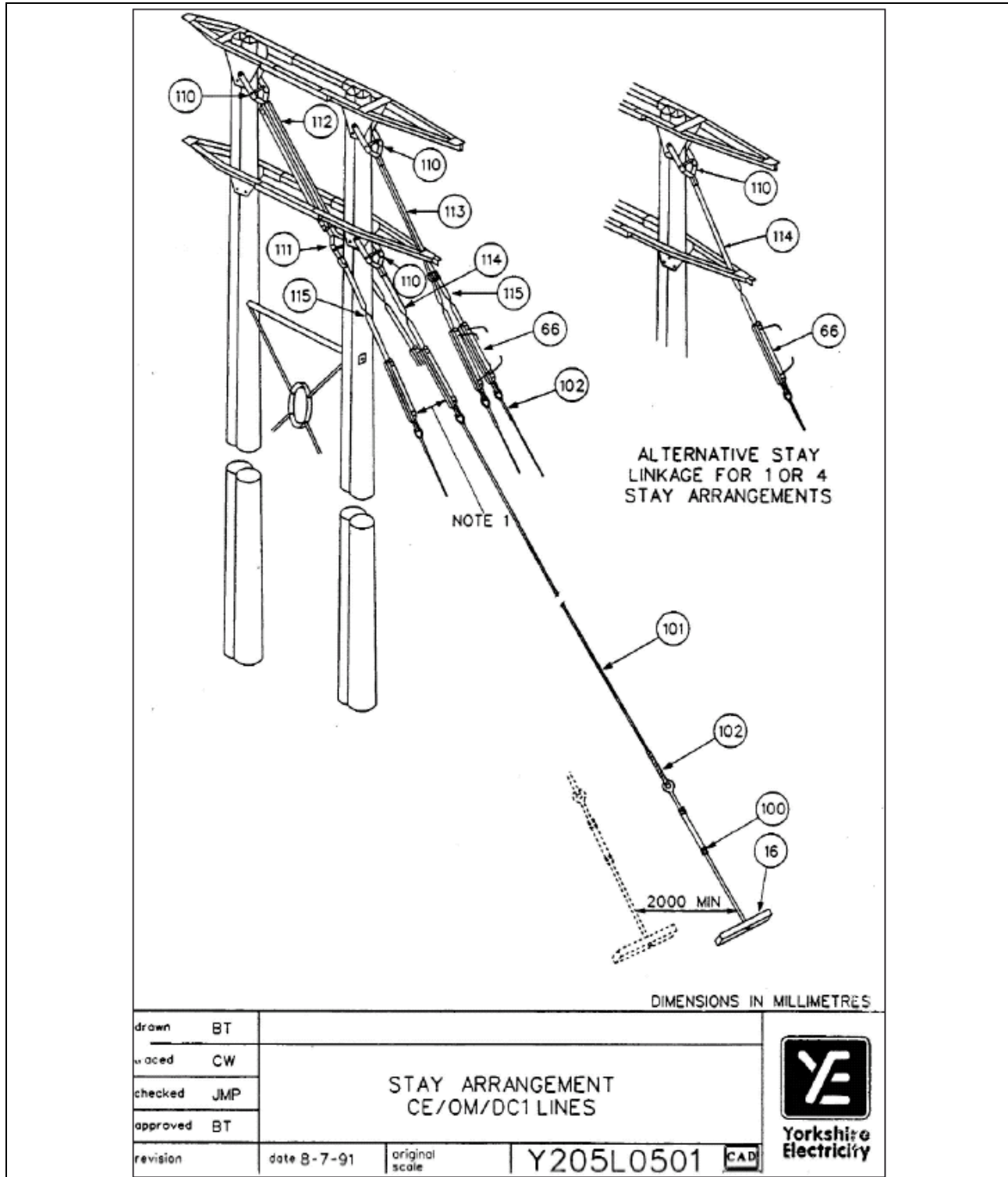
Appendix 3 – 11kV to 33kV Alternative Stay Arrangement

Alternative arrangements to drawing 1000439108 (for materials see appendix 2 above)



Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	17	of	27

Appendix 4 - Typical CE/OM/DC1 Stay Arrangement



Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	18	of	27

Materials For Y205L0501

Item	No	Cat	Description	Drawing
16	1	235124	Kicking /Stay Block (Type2)	1000439103
66	1	346596	Beam Insulator	109.101.0372 sht2
100	1	231477	Stay Rod (Type 2)	1000439101
101		229583	Stay Wire (7/4.00mm) Grade 1150	
102	3	255217	Stay Grip	1091010541

Stay Linkage – Top Crossarm, Outer Limb, Single Stay

110	1		Stay Link (type A)	
114	1		Stay Strap (Type 2) long	

Stay Linkage – Top Crossarm, Outer Limb, Double Stay

110	1		Stay Link (type A)	Y003X3213
113	1		Stay Strap (Type 1)	Y003X3212
115	2		Stay Strap (Type 3)	Y003X3212

Stay Linkage – Bottom Crossarm, Outer Limb, Single Stay

110	1		Stay Link (type A)	Y003X3213
114	1		Stay Strap (Type 2) Short	Y003X3212

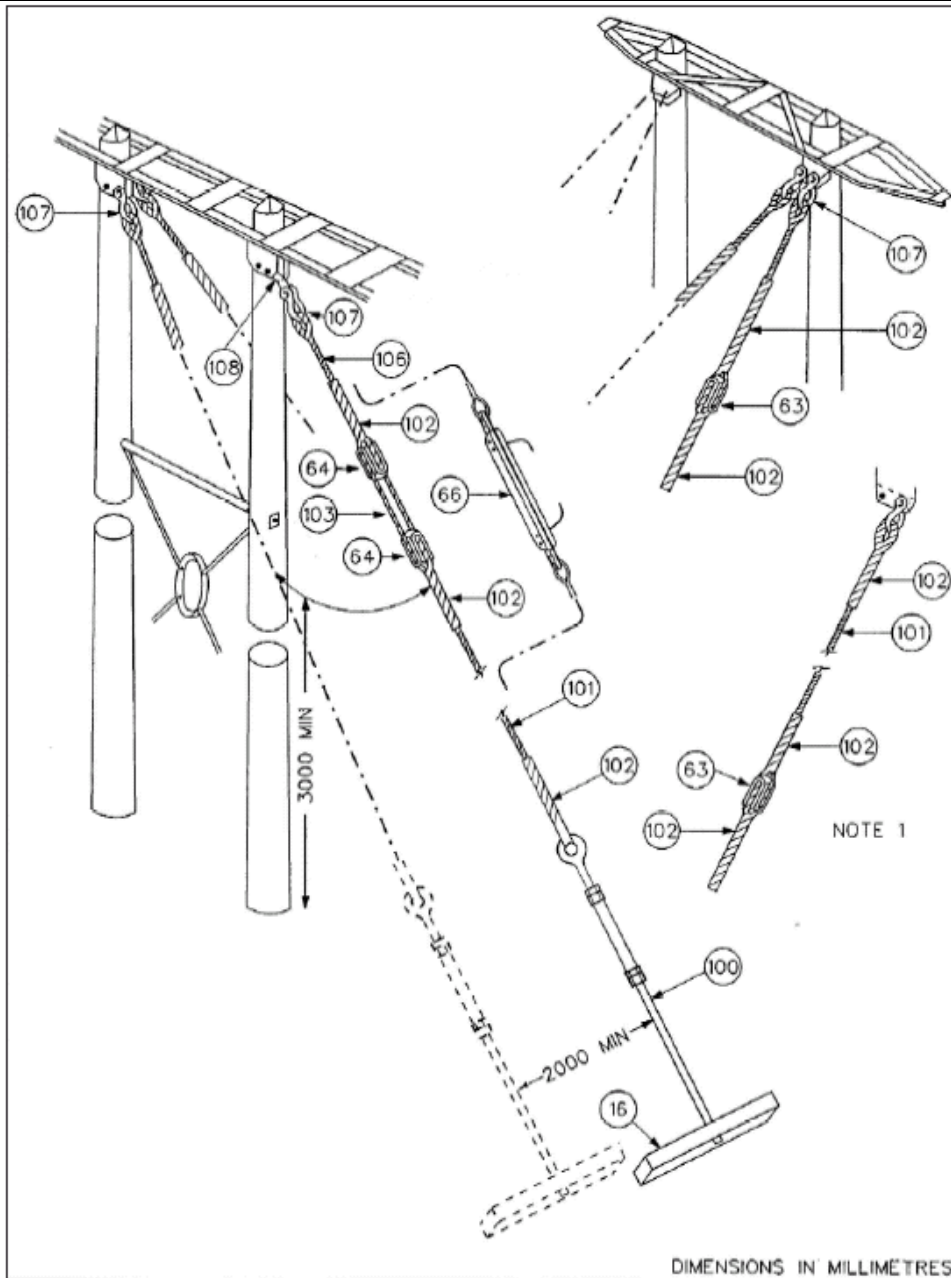
Stay Linkage – Bottom Crossarm, Outer Limb, Single Stay

110	1		Stay Link (type A)	Y003X3213
111	1		Stay Link (type B)	Y003X3213
112	1		Stay Insulating Insert	Y003X3211
115	2		Stay Strap (Type 3) Long	Y003X3212

Arcing horns not required on lower stays

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	19	of	27

Appendix 5 – Typical CE/C/37 Stay Arrangement



drawn	BT	STAY ARRANGEMENT CE/C/37 LINES		 Yorkshire Electricity	
traced	JEF				
checked	JMG				
approved	BT				
revision	date	13-6-91	original scale	Y204L0501	CAD

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	20	of	27

Materials For Y204L0501

Item	No	Cat	Description	Drawing
16	1	235124	Kicking /Stay Block (Type2)	1000439103
100	1	231477	Stay Rod (Type 2)	1000439101
101		229583	Stay Wire (7/4.00mm) Grade 1150	
102	3	255217	Stay Grip	1091010541
106	1	225367	Shackle Stay make Off	Y003X3205
107	1	226365	Shackle	Y003X3205
108	1	220343	Stay Bracket	Y003X3214
109	1	244922	Stay Thimble Type 1	1000439109

Stay Insulator Components – unearthed 33kV

64	2	253743	Stay Insualtor (Type 2)	1000439107
103	1	260183	Stay Link	1000439107 sht3

Stay Insulator Components – unearthed 66kV

66	1	346596	Beam Insulator	109.101.0372 sht2
----	---	--------	----------------	-------------------

Stay Insulator Components – earthed 33kV & 66kV

63	1	248232	Stay Insulator (Type 1)	1000439107
----	---	--------	-------------------------	------------

Extended stay tops – for use on structures with auxiliary equipment

Additional materials

101		216606	Stay Wire (7/4.00mm) Grade 1150	
102	1	255217	Stay Grip	1091010541

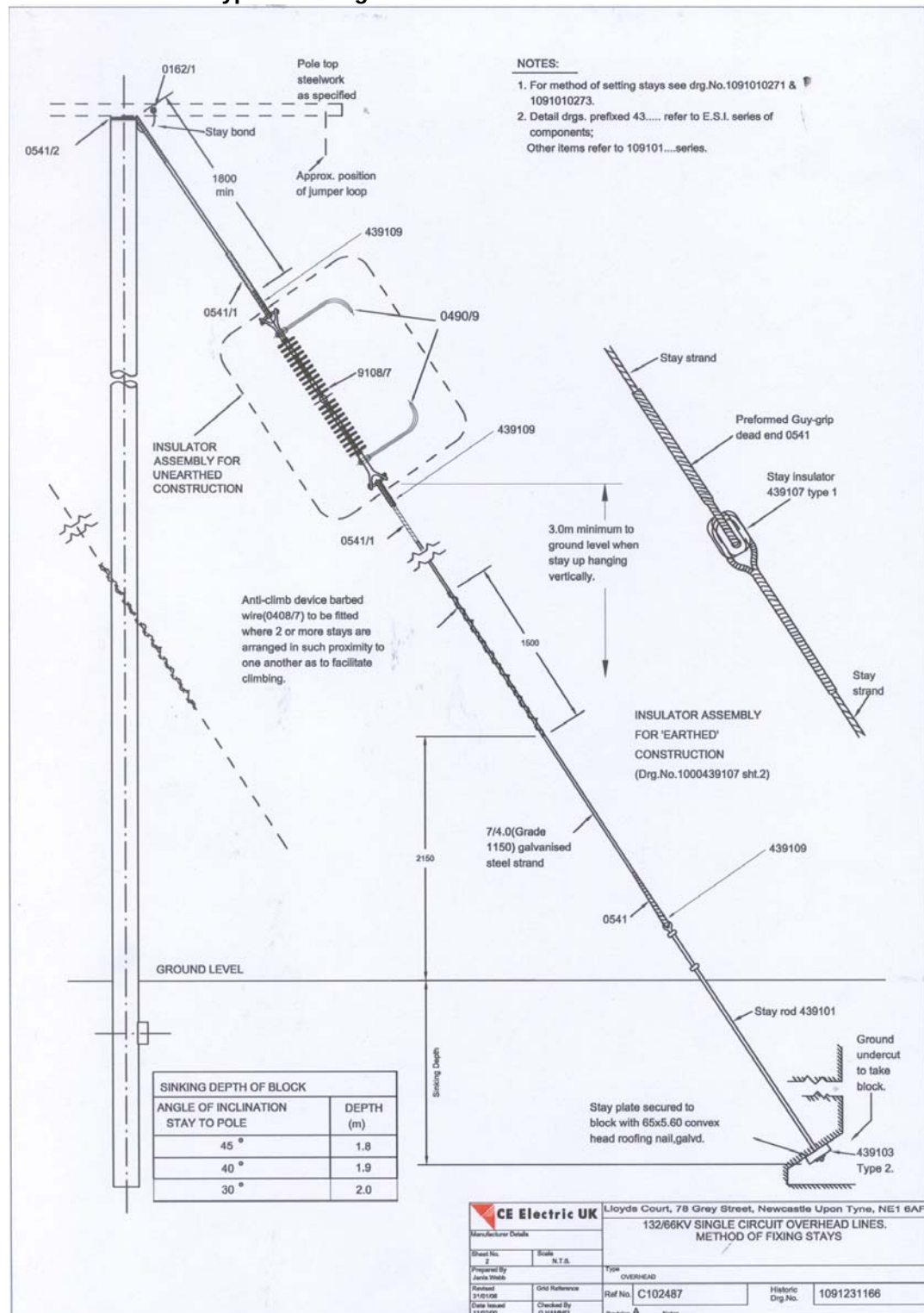
Stay Positions:-

On 1 or 2 stay installations the stays are fixed to the outer limb of the structure

On 3 stay installations 1 stay is fixed to the outer limb and 2 stays to the inner limb.

Document reference	NSP/004/104						
Version:-	3.0	Date of Issue:-	June 2010	Page	21	of	27

Appendix 6 – 66kV/132kV Typical Arrangement



Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	22	of	27

Materials for 1091231166 sht2

Item	No	Cat	Description	Drawing
16	1	235124	Kicking /Stay Block (Type2)	1000439103
100	1	231477	Stay Rod (Type 2)	1000439101
101		229583	Stay Wire (7/4.00mm) Grade 1150	
102	3	255217	Stay Grip	1091010541
-	3	244922	Stay Thimble	1000439109
Pole Top Terminations				
105	1	225342	Pole top make off	
104	1	226738	Stay Splice	
101		229583	Stay Wire (7/4.00mm) Grade 1150	
Or				
-	1	250209	Pole top dead end	1000439108 sht6
101		229583	Stay Wire (7/4.00mm) Grade 1150	

Insulator requirements for:-

66 or 132kV earthed arrangements

63	1	248232	Stay Insulator (Type 1)	1000439107
----	---	--------	-------------------------	------------

66 or 132kV unearthed arrangements

	1	346596	Stay Insulator	109.101.0372 sht2
			Stay Insulator Arcing Horns	1091010490 item 9

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	23	of	27

Appendix 7 – Stay Make offs

ITEM 1

STAY PLATE FIXING
SINGLE STAY

ITEM 2

STAY PLATE FIXING
TWO STAYS

NOTES:-

1. DETAIL DRAWINGS PREFIXED 43..... REFER TO ESI BASED DRAWINGS
OTHER DETAIL DRAWINGS REFER TO 1 09 101 SERIES.
2. BOND WIRE TO BE CENTRE WIRE OF 7 / 4.0 STAY STRAND.
3. STAYPOLE TO UTILISE LOWER ANTI - SPLIT BOLT. WHERE TWO STAYS
TO BE FITTED THE UPPERMOST ANTI - SPLIT BOLT SHALL BE USED
FOR THE SECOND STAY ASSEMBLY

ALL DIMENSIONS SHOWN ARE IN MILLIMETRES UNLESS OTHERWISE STATED

CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		OVERHEAD LINES H.V. CONSTRUCTION POLE TOP STAY MAKE - OFFS	
Drawn No.	Scale	Type	
4	N.T.S.	ESI	
Prepared By	Drawn By	Ref No.	Historic
James Wright	015480	C733982	1 00 043 9108
Checked By	Drawn By	Revision	Notes
G. HAMMILL	015480	E	

Document reference	NSP/004/104				
Version:-	3.0	Date of Issue:-	June 2010	Page	24 of 27

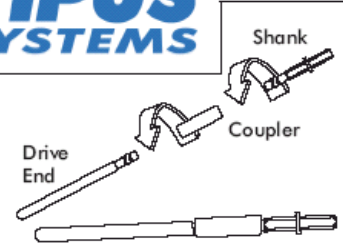
Appendix 8 - Platipus Utility System



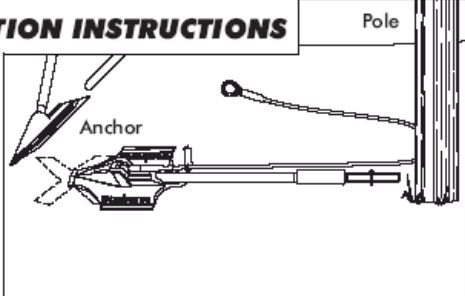
CAUTION!
All underground services must be located prior to installation of an anchor. Use two people to lift the hydraulic hammer.

SAFETY!
Wear safety clothing, especially ear defenders when using the hydraulic hammer.

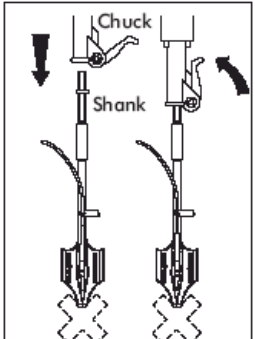
INSTALLATION INSTRUCTIONS



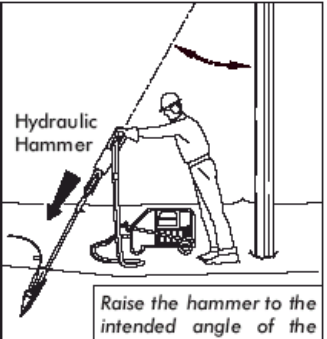
Connect the shank and drive end using the coupler. **NOTE: Drive rods, couplers and shank are left hand thread.**



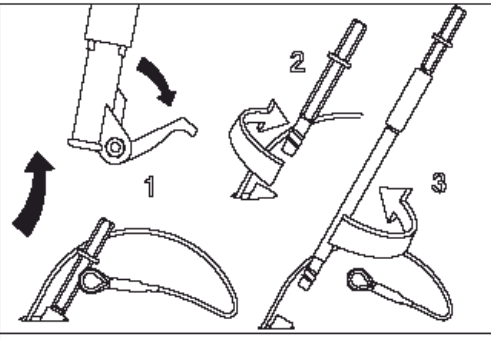
Insert the round tip of the drive end into the anchor and position the anchor at the point of entry into the ground facing away from the pole.



Insert the shank into the hammer chuck and lock in place.



Raise the hammer to the intended angle of the stay and begin driving the anchor into the ground at this angle until the top of the coupler just reaches ground level.



Lift the hammer from the shank. **DO NOT PULL THE DRIVE ROD UP OUT OF THE ANCHOR. RELOCATION MAY BE IMPOSSIBLE.** Attach another drive rod and coupler between the original drive end and shank.

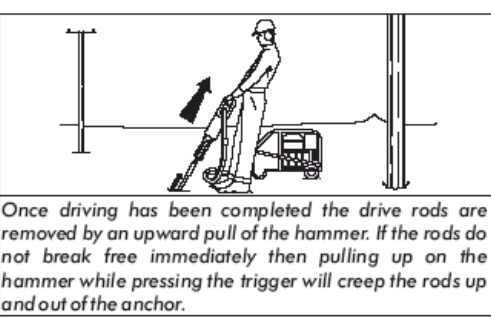
STOP



Replace the hammer on the shank and continue driving the anchor until the protective sleeving just reaches ground level.

CAUTION! DRIVING THE ANCHOR PAST THIS POINT CAN MAKE LOADLOCKING DIFFICULT.

NOTE: In reasonable ground the above process should not take more than 6-8 minutes. In very hard ground up to 20 minutes may be needed. In hard soil conditions a MINIMUM depth of 1.5 drive rod lengths must be reached.



Once driving has been completed the drive rods are removed by an upward pull of the hammer. If the rods do not break free immediately then pulling up on the hammer while pressing the trigger will creep the rods up and out of the anchor.



Remove the hammer from the shank.



Pull the drive rods out of the ground.

CAUTION! RODS AND COUPLERS MAY BE VERY HOT.

When the rods have been removed the anchor must be 'loadlocked' into its working position.

The PLATIPUS anchor must be **LOADLOCKED** and proof loaded before attaching the backstay. **LOADLOCKING INSTRUCTIONS ARE OVERLEAF.**

PLATIPUS ANCHORS LIMITED
 Kingsfield Business Centre, Philanthropic Road,
 REDHILL, Surrey, RH1 4DR, England.
 Tel: +44 (0) 1737 762300
 Fax: +44 (0) 1737 773395
 Web: www.platipus-anchors.com
 E-Mail: info@platipus-anchors.com

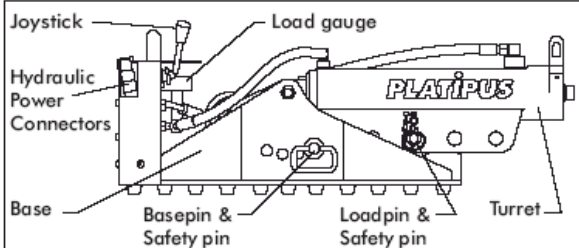
210803 CRF281

CAUTION! - This document may be out of date if printed

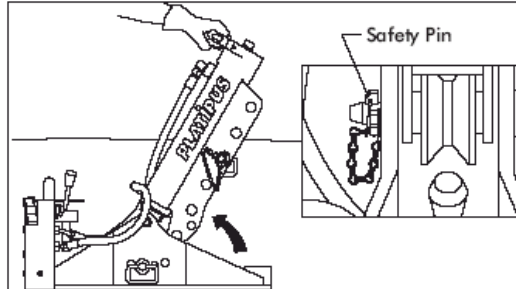
Document reference	NSP/004/104				
Version:-	3.0	Date of Issue:-	June 2010	Page	25 of 27

LOADLOCKING INSTRUCTIONS

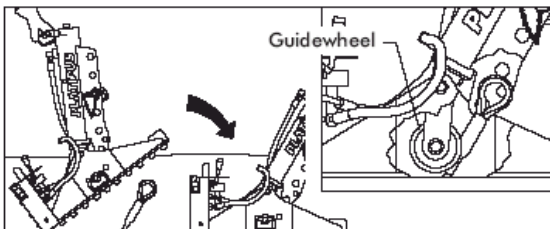
THE LOADLOCKING PROCEDURE IS TO BE PERFORMED BY ONE OPERATOR ONLY.



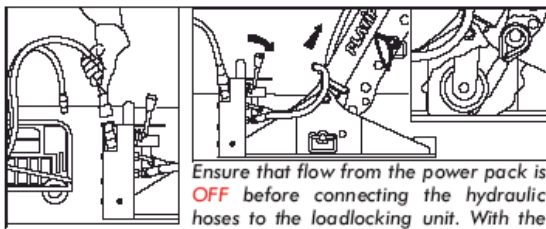
The SJ2 hydraulic powered loadlocking unit in the 'stowed' position. **CAUTION!! LOADING THE UNIT IN THIS POSITION WILL SEVERELY DAMAGE IT.**



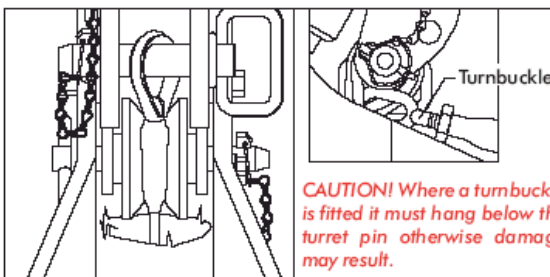
Remove the basepin from the loadlocking unit base and lift the turret to the intended angle of the backstay. Replace the basepin through the base and the wheel to lock the turret in position. **ENSURE THAT THE SAFETY PIN IS SECURELY ON THE BASEPIN.**



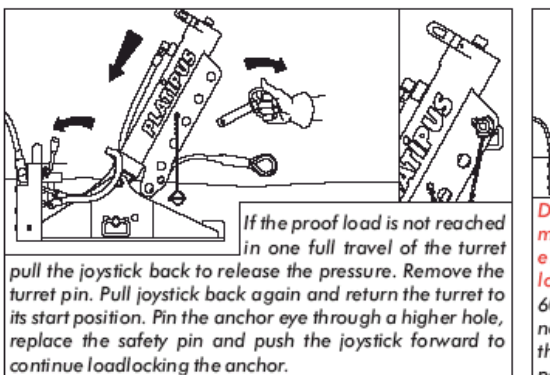
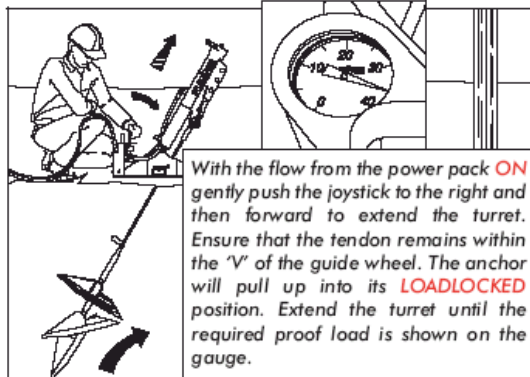
Place the loadlocking unit over the anchor eye with the turret pointing to the pole and the tendon in front of the guide wheel.



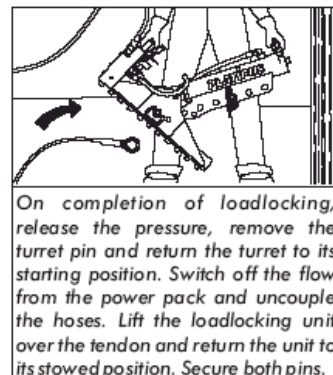
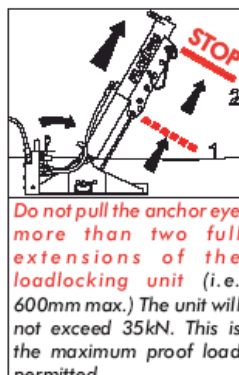
Ensure that flow from the power pack is **OFF** before connecting the hydraulic hoses to the loadlocking unit. With the flow **ON** the turret can be inched up to align the lowest available hole with the anchor eye by pushing the joystick to the right and then forward a small amount.



Push the turret pin through the lowest available hole and engage the anchor eye. **Fit the safety pin. Ensure that the anchor tendon sits in the 'V' of the guide wheel.**



7a **CONTACT PLATIPUS ANCHORS LTD IMMEDIATELY IF ANY COMPONENT IS MISSING OR DAMAGED**



Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	26	of	27

Appendix 9 – Installation Instructions for Augured Ground Anchor Systems

Choice of Anchor

Different types of soil affect the holding capability of the screw anchor and this should be considered prior to installation. As a guide the ground type can usually be determined when inspecting the backfill material from newly excavated adjacent pole holes or from local experience. The standard CE Electric UK stock items are listed below but additional specialist anchors are available for non-standard ground conditions.

224881	10" Single Helix Assembly
224899	12" Single Helix Assembly
217448	Barrel Turnbuckle c/w Eye, Thimble & Locknut
226175	Rod 2012mm (7 ft)
222414	Rod Coupling
226183	Rod 1050mm (4ft)

Generally the larger the disc diameter and number of discs the stronger the holding strength of the stay assembly. In areas where the ground type is peat or fine sand twin helical discs are available. Alternately the holding strength can be increased by adding a 1050mm extension to the standard 2012mm rod.

Installing the Auger Stay Anchor

Before commencing work ensure that utilities are identified and safe dig practices are implemented.

Note – In view of the height of the machinery when installing the screw anchor extreme care should be taken when working in the proximity of live overhead line conductors.

- Mark out the position of the stays as detailed in the line schedule.
- Insert the helical assembly into the hydraulic drive attachment and ensure it is locked into place with the holding bolts.
- The aim is to install the anchor in a smooth progressive manor ensuring that it remains aligned thus reducing the ground disturbance and increasing strength.
- During the anchor installation the driver operator must monitor the applied torque on the gauge attached to the machinery. If the value exceeds 7000Nm the body of the rod twists and the anchor could be damaged.
- Drill the first 0.3m of the anchor into the ground in the vertical position.
- Set the anchor to the required angle and continue to install until the appropriate depth is achieved. The minimum size rod installed shall be 2012mm with no more than 300mm protruding out of the ground.
- In the final stages of installation the reading of the torque gauge should be noted by the operator.
- When the installation is completed withdraw the drive attachment from the ground after firstly ensuring that the bolts holding the anchor in place are removed.

Determining the Holding Capacity

The operator shall look to achieve a final installation torque of between 2000Nm and 7000Nm.

The strength can be increased by changing the rod length, disc diameter or number of discs.

If the final torque achieved is less than 2000Nm it is likely that insufficient holding strength has been achieved. If this cannot be increased by changing the assembly type a standard rod and block system

Document reference		NSP/004/104					
Version:-	3.0	Date of Issue:-	June 2010	Page	27	of	27

should be installed. Anchors can be unscrewed if they fail to meet the required torque. Install any new anchor a minimum of three helix diameters away from the original to ensure that it is installed in undisturbed ground.

Completing the installation

When the installation is completed and the drive attachment removed the barrel turnbuckle can be attached as a future means of adjustment. Any disturbed ground shall be consolidated.